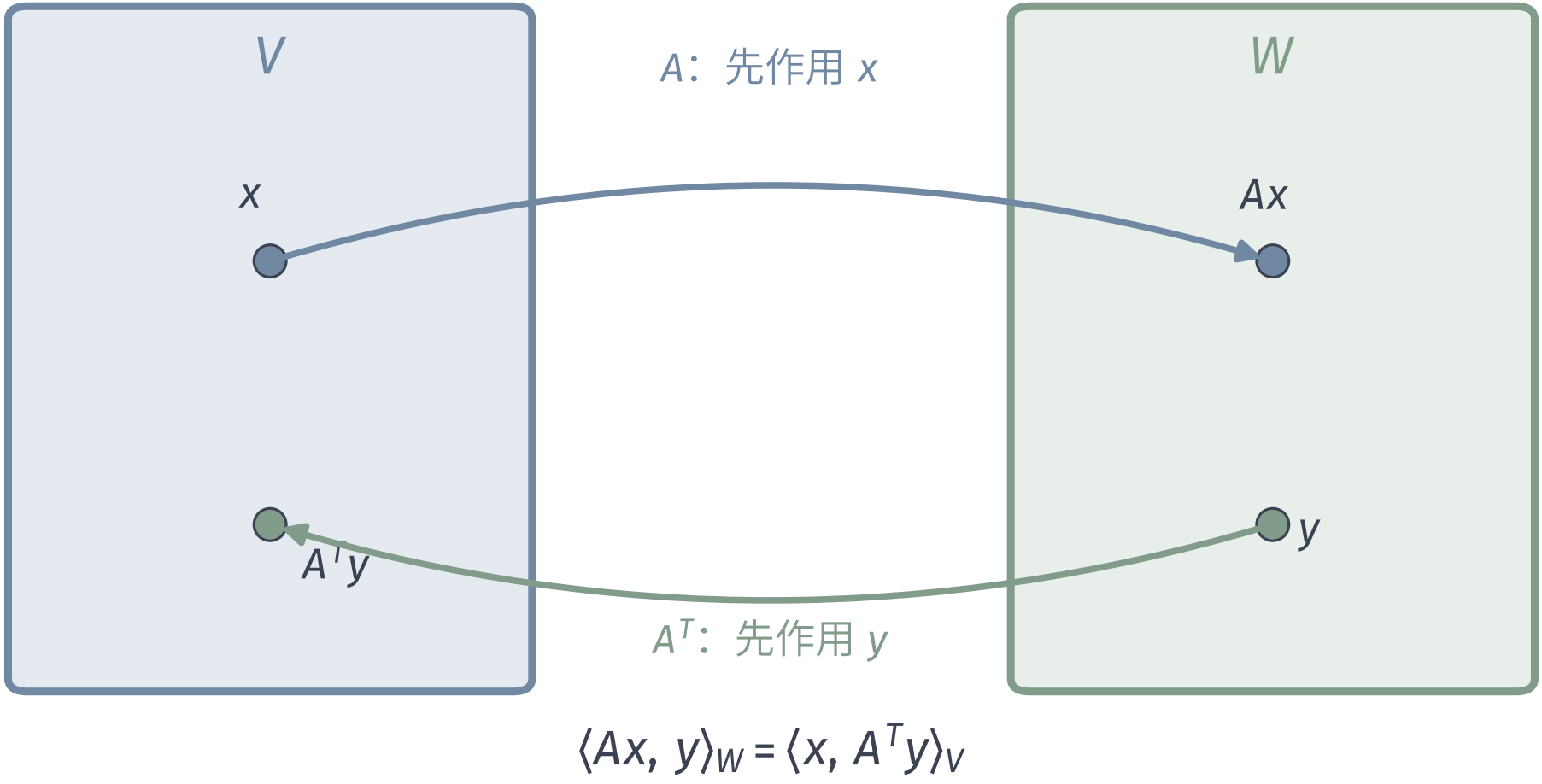

$$\langle Ax,\, y\rangle_W = \langle x,\, A^T y\rangle_V$$